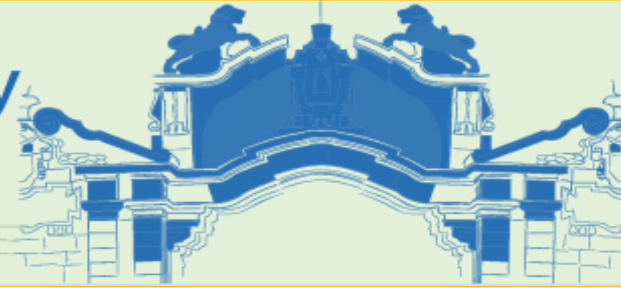




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Addis Ababa Institute of Technology

School of Mechanical and Industrial Engineering

Engineering Materials I

(MEng 2122)

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June, 2022

Chapter - 5

Diffusion in Solids

Diffusion in Solids

Overview Application of Diffusion in Mechanical Structures

Carbon diffusion in surface treatment.



Why Study Diffusion?

The reason why we study the term diffusion is primarily to answer the following basic questions:

- ❖ What does it mean diffusion in general?
- ❖ What is diffusion in solids and its principle?
- ❖ What are the mechanisms of diffusion? /How diffusion occurs?/
- ❖ Why diffusion is important in materials processing and treatment?
- ❖ What are the basic factors that influence diffusion?
- ❖ How can the rate of diffusion be predicted for some simple cases?
- ❖ How does diffusion depend on structure and temperature?
- ❖ What real environmental application diffusion has?

Learning Objectives

After completing this chapter, you should be able to:

- ☐ List and describe the two atomic mechanisms of diffusion.
- ☐ Distinguish between steady-state and non-steady state diffusion.
- ☐ Write *Fick's* first and second laws in equation form and define all parameters.
- ☐ Note the kind of diffusion for which each of these equations is normally applied.
- ☐ Write the solution to *Fick's* second law for diffusion into a semi-infinite solid when the concentration of diffusing species at the surface is held constant. Define all parameters in this equation.
- ☐ Calculate the diffusion coefficient for a material at a specified temperature, given the appropriate diffusion constants.

Issues to address

- How does diffusion occur?
- Why is it an important part of processing?
- How can the rate of diffusion be predicted for some simple cases?
- How does diffusion depend on structure and temperature?

Introduction to Diffusion

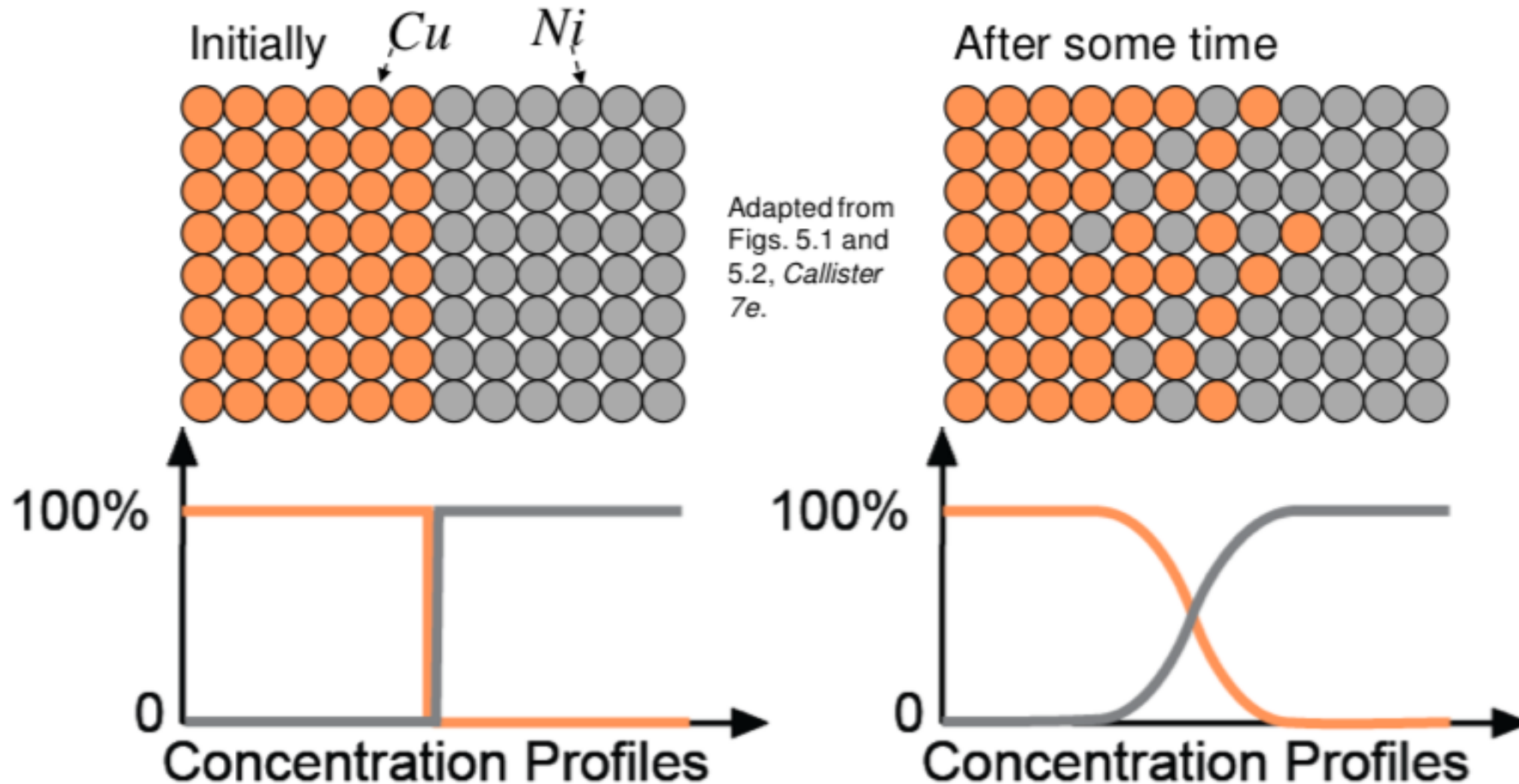
- **Diffusion** is the phenomenon of material transport by atomic motion.

Diffusion → mass transport by atomic motion.

- **Mechanisms**: This transfer may be either within a **specific solid** or from a **liquid**, a **gas**, or **another solid** phase.
- This is accomplished by **diffusion**.
 - Gases & Liquids – **random** (Brownian) motion
 - Solids – vacancy diffusion or interstitial diffusion
- The **process** by which atoms of one metal **diffuse** into another is termed **interdiffusion**, or **impurity diffusion**.
- There is a net drift or transport of atoms from high- to low-concentration regions.

Introduction cont.

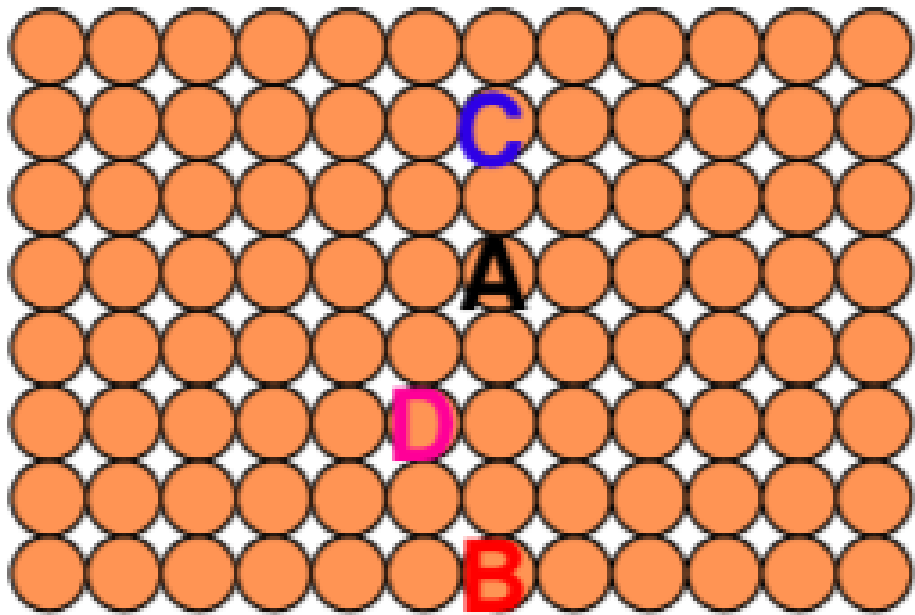
- **Interdiffusion:** In an alloy, atoms tend to migrate from regions of high conc. to regions of low conc.



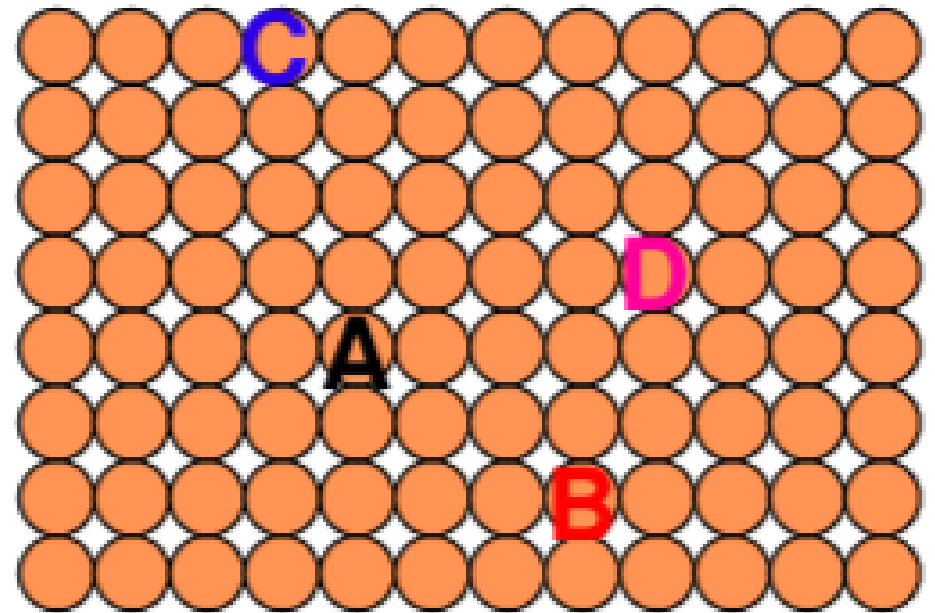
Introduction cont.

Self-diffusion: In an elemental solid, atoms also migrate.

Label some atoms



After some time



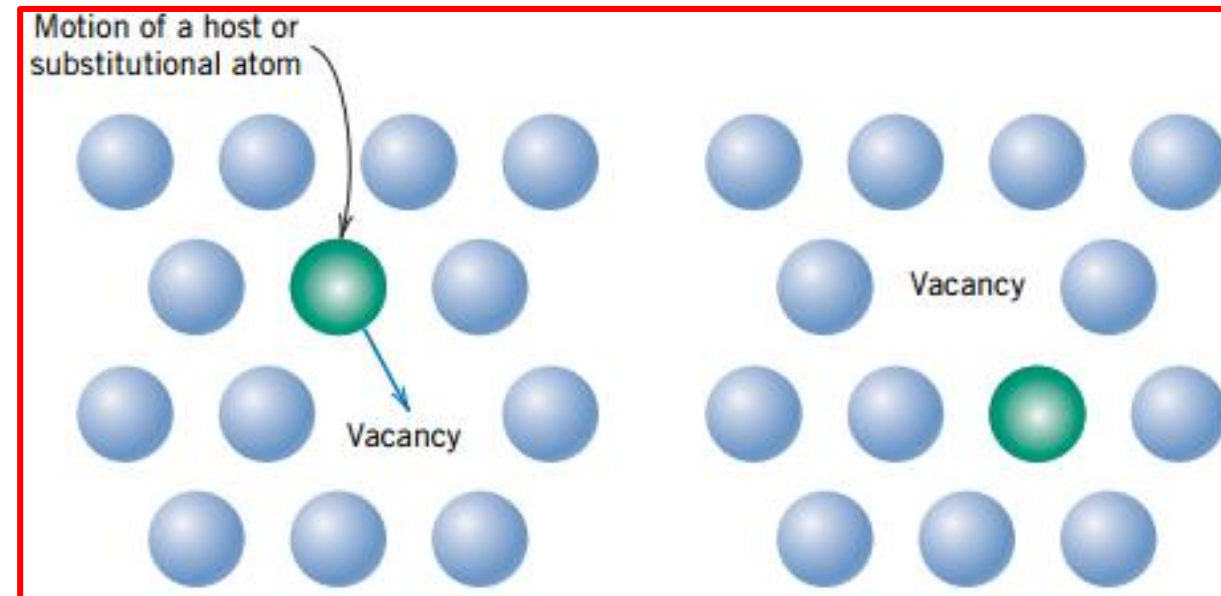
Diffusion Mechanisms

- **Diffusion** is just the stepwise migration of atoms from lattice site to lattice site. → from atomic perspective.
- The atoms in solid materials are in constant motion, rapidly changing positions.
- For the **movement of an atom**, two conditions must be met:
 - 1) there must be an empty adjacent site, and
 - 2) the atom must have sufficient energy to break bonds.
- The fraction of **diffusion increases** with rising temperature.
- **Two possible diffusion mechanisms:**
 - Vacancy diffusion
 - Interstitial diffusion

Diffusion Mechanisms

Vacancy diffusion:

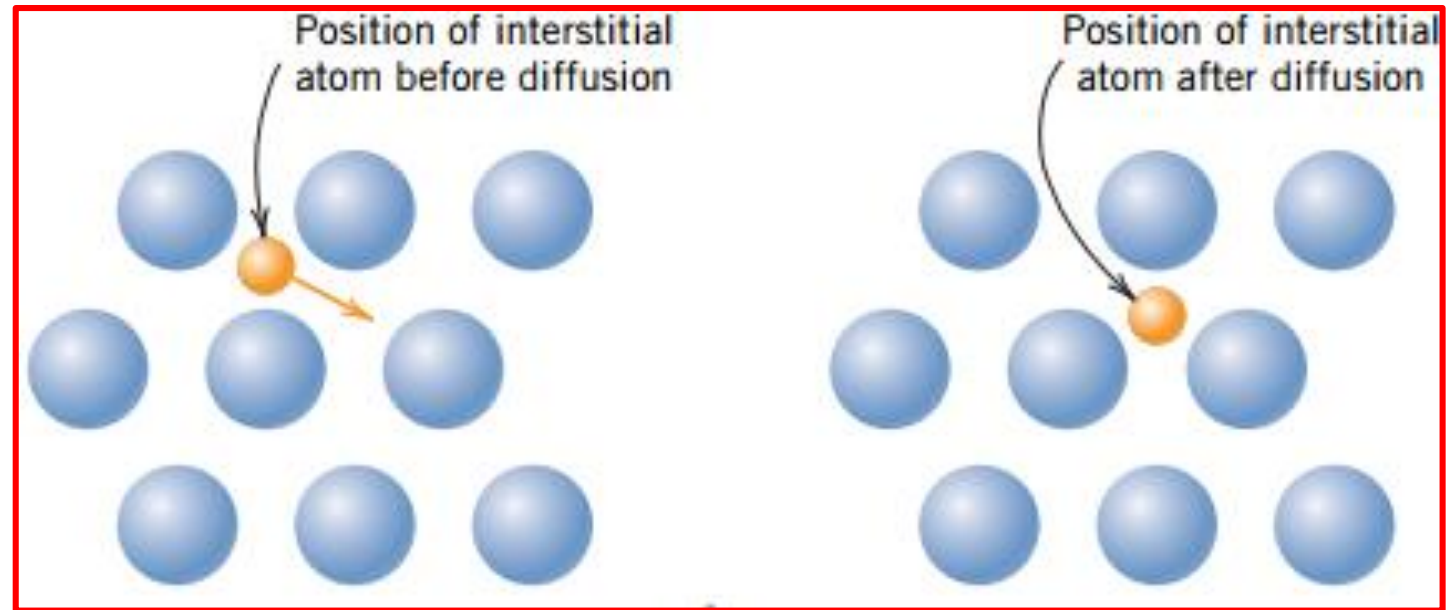
- It is the **interchange** of an atom from a normal lattice position to an adjacent vacant lattice site or vacancy.
- Vacancy diffusion – needs presence of vacancies.
- The extent to which *vacancy diffusion* can occur is a function of the number of these defects.
- The **diffusion** of atoms in one direction corresponds to the motion of vacancies in the opposite direction.
- Significant concentrations of vacancies may exist in metals at elevated temperatures.



Diffusion Mechanisms

Interstitial diffusion:

- It is the migration of atoms from an *interstitial position* to a neighboring one that is empty.
- This mechanism is found for **interdiffusion of impurities** such as H, C, N, and O, which have small atoms.
- **Host** or **substitutional** impurity atoms rarely form interstitials and do not diffuse by this mechanism → termed **interstitial diffusion**.



Principle of Diffusion

- Diffusion is a *time-dependent* process.
- **Diffusion flux (J)**: is the *rate of mass transfer*.
- It is the **mass or number** of atoms diffusing through a *unit area* per *unit time*.
- Its mathematical expression is;

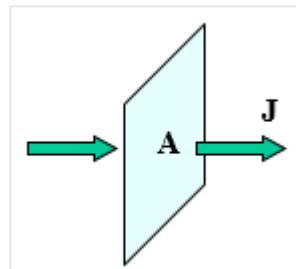
$$J = \frac{M}{At}$$

M = Diffusing mass (kg)

A = cross sectional area of diffusion (m^2)

t = time of diffusion (s).

- The unit of J is $\text{kg}/\text{m}^2\text{s}$.



- Fick's first law** — diffusion flux for steady-state diffusion: *independent of time*
- This is considered as in **one direction**.
 - The mathematical expression of diffusion flux for steady-state diffusion is given by:

$$J = -D \frac{dC}{dx}$$

D = diffusion coefficient (m^2/s)

Minus sign means diffusion toward lower concentrations.

- J is proportional to the concentration gradient (dC/dx).

dC/dx is $\rightarrow$ a driving force or concentration gradient.

Fick's First Law

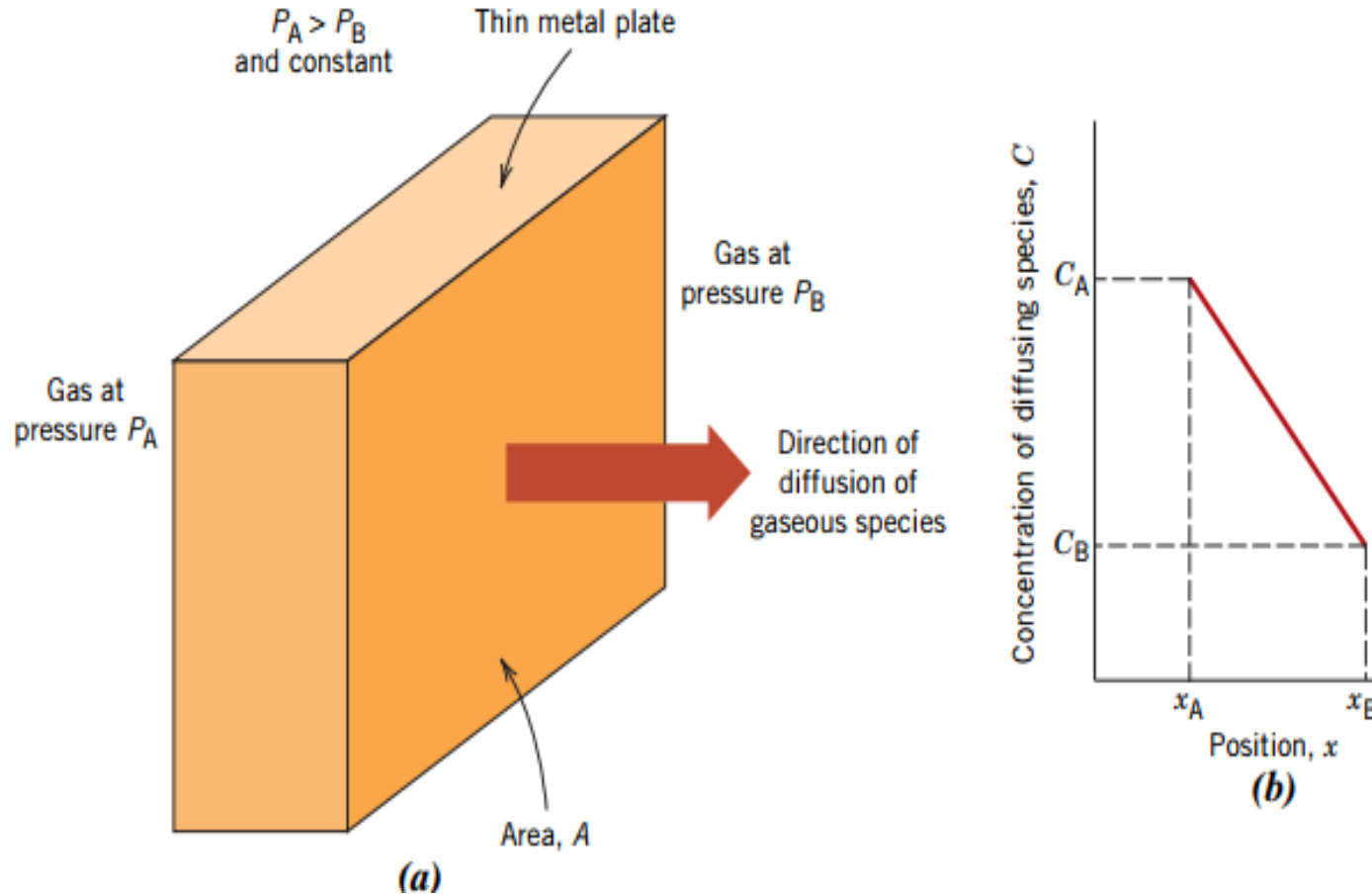


Figure: (a) Steady-state diffusion across a thin plate. (b) A linear concentration profile for the diffusion situation in (a).

- Fick's first law may be applied to diffusion of atoms of a gas through a thin metal plate. → concentrations of diffusing species on both surfaces held constant.
- **Steady-state diffusion** - the diffusion flux does not change with time.
- Concentration profile - is the curve plotted with concentration (C) vs position (x).

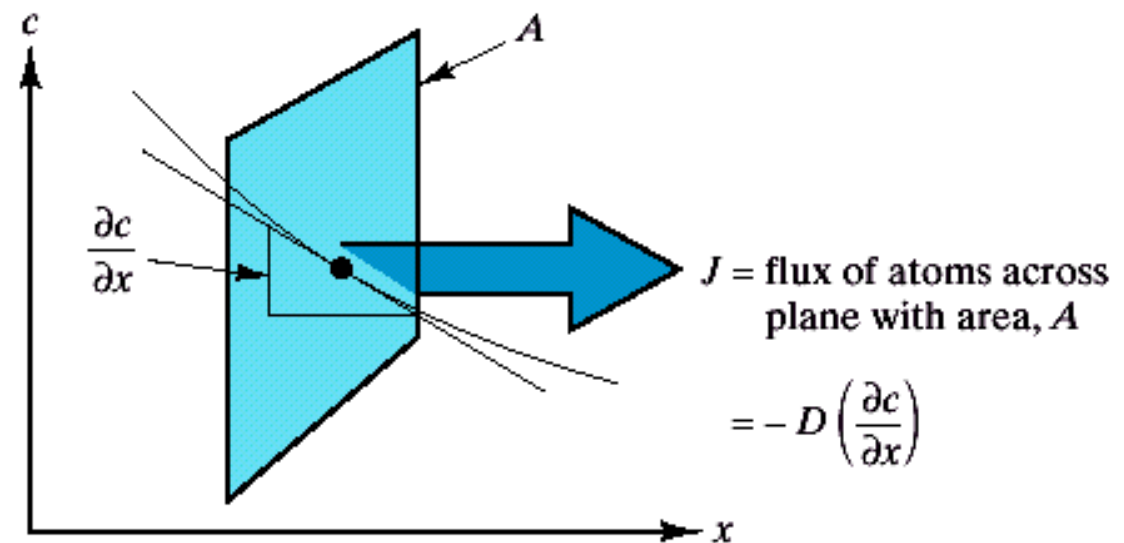
Fick's First Law

- The **slope** at a particular point of C-vs-x curve is called the **concentration gradient**.
- The **hydrogen** selectively diffuses through the sheet to the opposite side, which is maintained at a constant and lower **hydrogen** pressure.

C may be expressed in $\rightarrow \text{kg/m}^3$

$$\text{concentration gradient} = \frac{dC}{dx} = \frac{\Delta C}{\Delta x} = \frac{C_A - C_B}{x_A - x_B}$$

- Example of **steady-state diffusion** is found in the purification of **hydrogen gas**.
- One side of a thin sheet of palladium metal is exposed to the impure gas composed of **hydrogen** and other gaseous species such as **nitrogen**, **oxygen**, and water vapor.



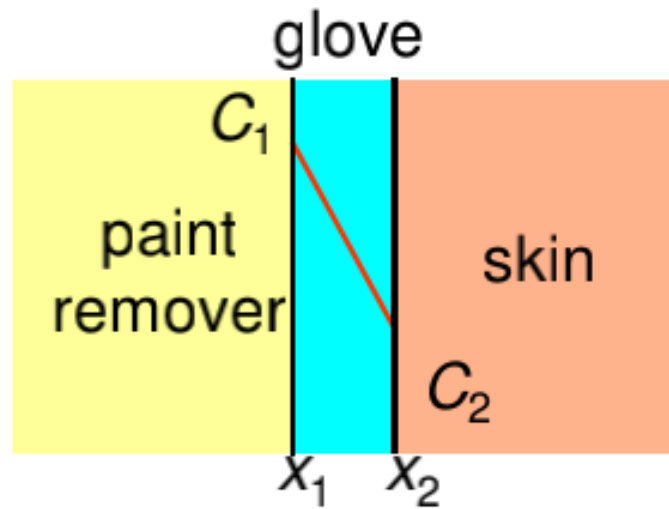
Example

1. Methylene chloride is a common ingredient of paint removers. Besides being an irritant, it also may be absorbed through skin. When using this paint remover, protective gloves should be worn.
 - If butyl rubber gloves (0.04 cm thick) are used, what is the diffusive flux of methylene chloride through the glove?

Data:

- diffusion coefficient in butyl rubber: $D = 110 \times 10^{-8} \text{ cm}^2/\text{s}$
- surface concentrations: $C_2 = 0.02 \text{ g/cm}^3$ $C_1 = 0.44 \text{ g/cm}^3$

Solution



$$J = -D \frac{dC}{dx} \cong -D \frac{C_2 - C_1}{x_2 - x_1}$$

Data: $D = 110 \times 10^{-8} \text{ cm}^2/\text{s}$

$$C_1 = 0.44 \text{ g/cm}^3$$

$$C_2 = 0.02 \text{ g/cm}^3$$

$$x_2 - x_1 = 0.04 \text{ cm}$$

$$J = -(110 \times 10^{-8} \text{ cm}^2/\text{s}) \frac{(0.02 \text{ g/cm}^3 - 0.44 \text{ g/cm}^3)}{(0.04 \text{ cm})} = 1.16 \times 10^{-5} \frac{\text{g}}{\text{cm}^2 \text{s}}$$

Fick's Second Law

Fick's Second Law – Non-steady State Diffusion

- **Non-steady state diffusion** occurs when the diffusion flux and the concentration gradient at some particular point in a solid is **varying with time**.
- Most practical diffusion situations are **non-steady state**.
- To see the **non-steady state diffusion** condition - concentration profiles at **three different diffusion times** (t_1 , t_2 , and t_3) are considered (see *figure 5.5*).

- **Fick's second law:**

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x} \left(D \frac{\partial C}{\partial x} \right)$$

- If the **diffusion coefficient (D)** is **independent** of composition, it can be simplified as;

$$\begin{aligned} \frac{\partial C}{\partial t} &= - \frac{\partial J}{\partial x} \\ &= D \frac{\partial^2 C}{\partial x^2} \end{aligned}$$

Fick's second law:
Diffusion equation
for **non-steady state**
diffusion (in one
direction)

Fick's Second Law

Non-steady State Diffusion

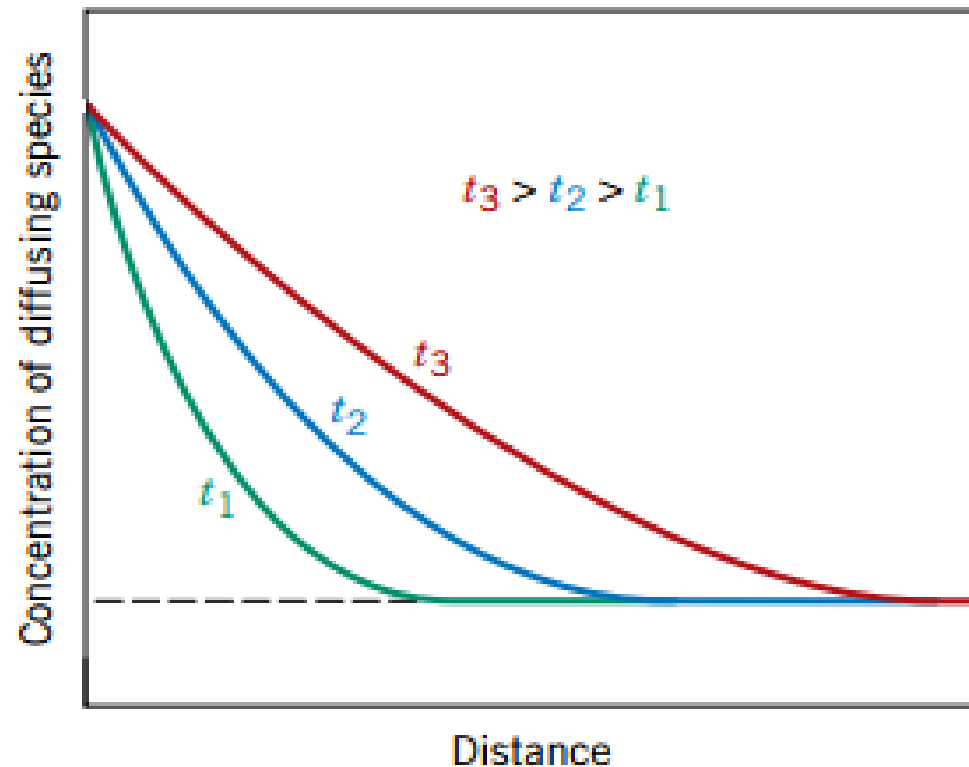


Figure 5.5: Concentration profiles for non-steady state diffusion taken at three different times, t_1 , t_2 , and t_3 .

- Atom needs enough thermal energy to break bonds and squeeze through its neighbors.

Energy needed $\rightarrow$ energy barrier $\rightarrow$

called the activation energy E_m (like Q)

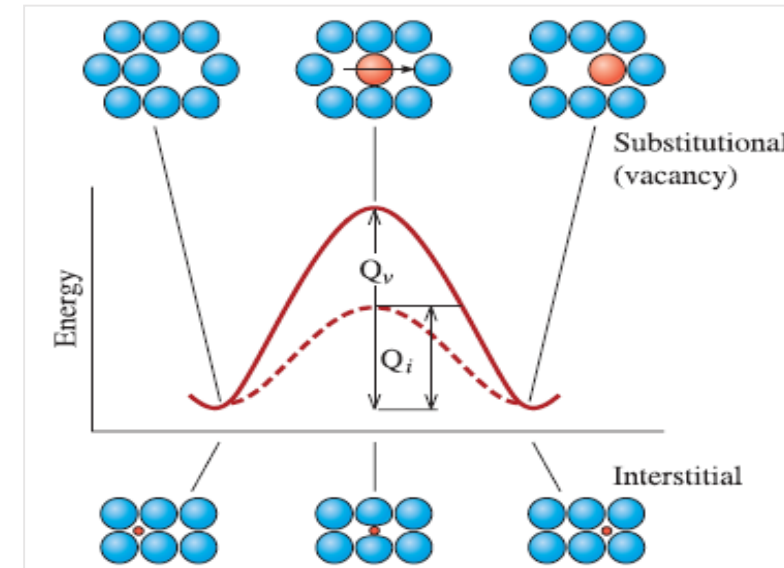


Diagram for vacancy and interstitial diffusion
Diffusion $\rightarrow$ thermally activated process

Fick's Second Law

- Solution to *Fick's second law* for the condition of **constant surface concentration** → for a semi-infinite solid:

$$\frac{C_x - C_0}{C_s - C_0} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right) \quad \text{..... (5.5)}$$

where C_x represents the concentration at depth x after time t .

The expression $\operatorname{erf}(x/2\sqrt{Dt})$ is the Gaussian error function.

$$\frac{C_1 - C_0}{C_s - C_0} = \text{constant} \quad \frac{x}{2\sqrt{Dt}} = \text{constant}$$

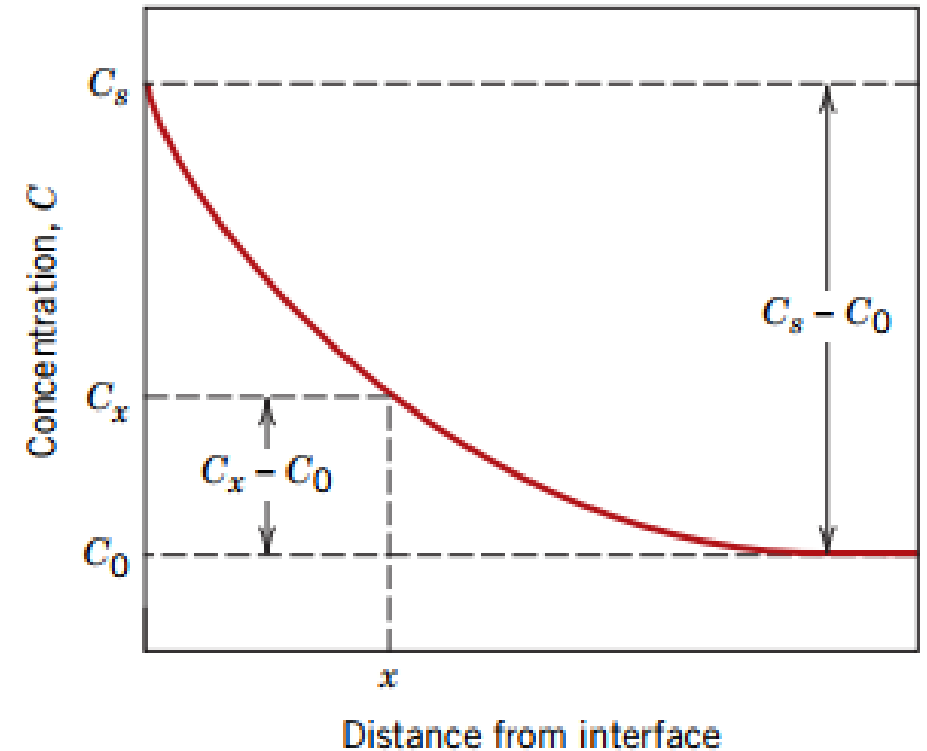


Figure 5.6: Concentration profile for non-steady state diffusion; concentration parameters

Factors Influencing Diffusion

■ Factors that affect diffusion:

- Diffusing species
- Temperature
- Microstructure
- Host material

Diffusing Species:

- The **magnitude** of the *diffusion coefficient* D is indicative of the rate at which atoms diffuse.
- The **diffusing** species and the host material influence the diffusion coefficient.
- **Self-diffusion** occurs by a **vacancy** mechanism, whereas **carbon diffusion** in iron is **interstitial**.

Temperature:

- Temperature has a profound influence on the **coefficients** and **diffusion rates**.
- The temperature dependence of the diffusion coefficients is:

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$$

Where, D_0 = a temperature-independent preexponential (m^2/s)

Q_d = the **activation energy** for diffusion (J/mol)

R = the gas constant, $8.31 \text{ J/mol}\cdot\text{K}$

T = absolute temperature (K)

Factors Influencing Diffusion

- The **activation energy** is the energy required to produce the diffusive motion of one mole of atoms.
- A **large activation energy** results in a relatively small diffusion coefficient.
- For evaluating D_0 and Q_d values:

$$\ln D = \ln D_0 - \frac{Q_d}{R} \left(\frac{1}{T} \right) \quad \text{or}$$

$$\log D = \log D_0 - \frac{Q_d}{2.3R} \left(\frac{1}{T} \right)$$

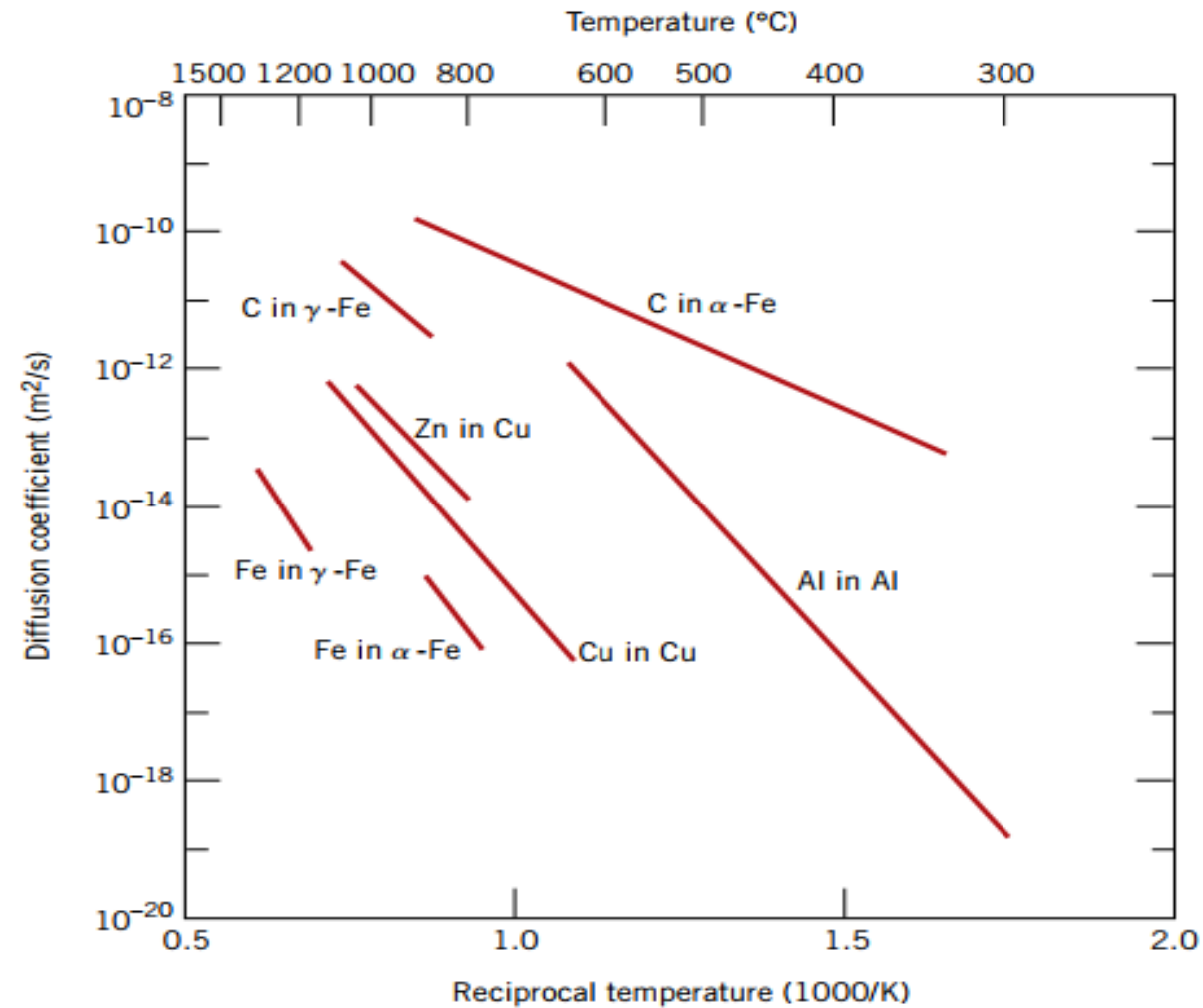
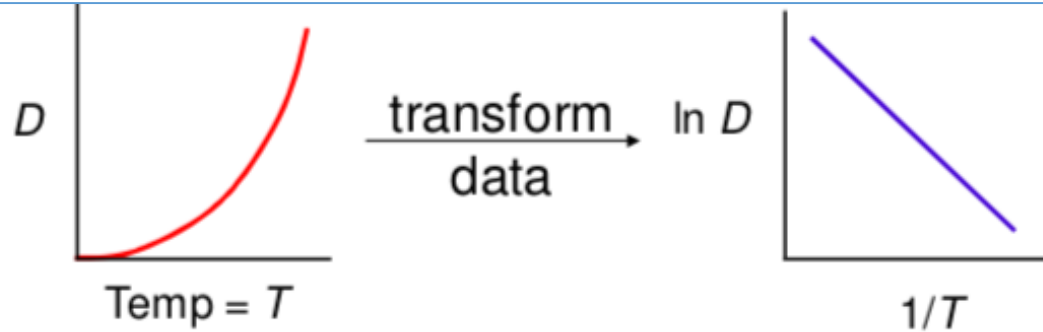


Figure 5.7: Plot of the logarithm of the diffusion coefficient versus the reciprocal of absolute temperature for several metals.

Example

- 1) If the diffusion coefficient (D) and activation energy (Q_d) for Cu in Si at 300°C are $D = 7.8 \times 10^{-11} \text{ m}^2/\text{s}$ and $Q_d = 41.5 \text{ kJ/mol}$. Then determine the diffusion coefficient at 350°C. [$R = 8.314 \text{ J/mol.K}$]

Solution



$$\ln D_2 = \ln D_0 - \frac{Q_d}{R} \left(\frac{1}{T_2} \right) \quad \text{and} \quad \ln D_1 = \ln D_0 - \frac{Q_d}{R} \left(\frac{1}{T_1} \right)$$
$$\therefore \ln D_2 - \ln D_1 = \ln \frac{D_2}{D_1} = -\frac{Q_d}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$D_2 = D_1 \exp \left[-\frac{Q_d}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right) \right]$$

$$T_1 = 273 + 300 = 573 \text{ K}$$

$$T_2 = 273 + 350 = 623 \text{ K}$$

$$D_2 = (7.8 \times 10^{-11} \text{ m}^2/\text{s}) \exp \left[\frac{-41,500 \text{ J/mol}}{8.314 \text{ J/mol} \cdot \text{K}} \left(\frac{1}{623 \text{ K}} - \frac{1}{573 \text{ K}} \right) \right]$$

$$D_2 = 15.7 \times 10^{-11} \text{ m}^2/\text{s}$$

Diffusion in Semi-conducting Materials

- Solid-state diffusion is the fabrication of semiconductor integrated circuits (ICs).
- IC chip is a thin square wafer having dimensions on the order of $6 \times 6 \times 0.4$ mm.
- Millions of interconnected electronic devices and circuits are embedded in the chip faces.
- Single-crystal silicon is the base material for most ICs.
- For an efficient ICs - very precise concentrations of impurity must be incorporated.
- This silicon chip pattern formation is accomplished by **atomic diffusion**.
- Two heat treatments are used in this process.
 - 1) **Pre-deposition step**, impurity atoms are diffused into the silicon. From 900°C and 1000°C for $< 1\text{h}$. By eq. 5.5.
 - 2) **Drive-in diffusion**, is used to transport impurity atoms farther into the silicon in order to provide a more suitable concentration distribution without increasing the overall impurity content.

Diffusion in Semi-conducting Materials

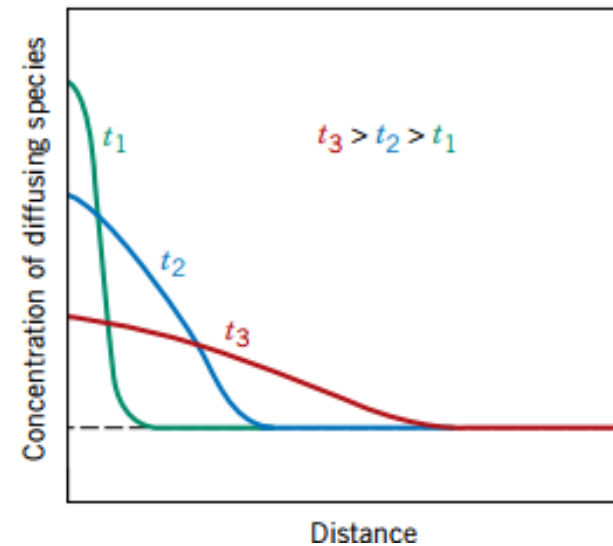
- The *drive-in diffusion* treatment is carried out up to 1200°C in an oxidizing atmosphere.
- The solution to *Fick's second law* for *drive-in diffusion* takes the form:

$$C(x,t) = \frac{Q_0}{\sqrt{\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right)$$

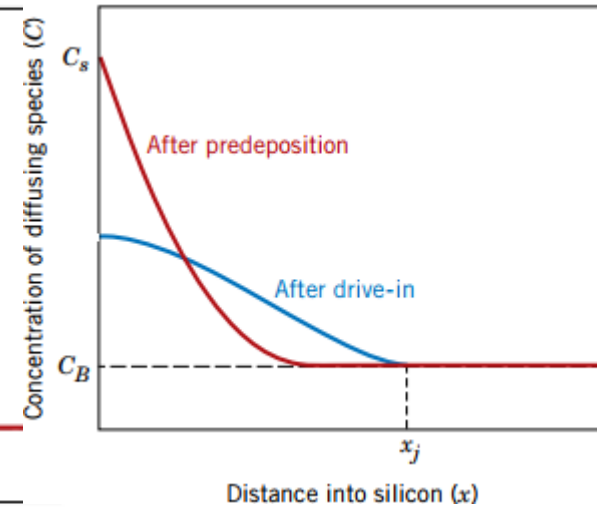
Where, Q_0 = total amount of impurities in the solid → introduced during predeposition treatment.

$$Q_0 = 2C_s \sqrt{\frac{D_p t_p}{\pi}}$$

where C_s = the surface concentration for the predeposition step, D_p = the diffusion coefficient, and t_p = the predeposition treatment time.



Drive-in diffusion of semiconductors at three different times, t_1 , t_2 , and t_3 .



Profiles after predeposition and drive-in diffusion treatments for semiconductors.

Other Diffusion Paths

- Atomic migration may also occur along:
 - dislocations,
 - grain boundaries, and
 - external surfaces.
- These are sometimes called *short-circuit diffusion paths* in as much as rates are much faster than for bulk diffusion. However, in most situations short_circuit contributions to the overall diffusion flux are insignificant because the cross_sectional areas of these paths are extremely small.

Summary

- **Solid-state diffusion** is a means of mass transport within solid materials by stepwise atomic motion.
- The term *interdiffusion* refers to the migration of impurity atoms; for host atoms, the term *self-diffusion* is used.
- **Two mechanisms** for diffusion are possible: vacancy and interstitial.
 - *Vacancy diffusion* occurs via the exchange of an atom residing on a normal lattice site with an adjacent vacancy.
 - *Interstitial diffusion*, an atom migrates from one interstitial position to an empty adjacent one.
- For a given host metal, interstitial atomic species generally diffuse more rapidly.

Summary

- *Diffusion flux* is defined in terms of mass of diffusing species, cross-sectional area, and time.
- *Diffusion flux* is proportional to the negative of the concentration gradient according to Fick's first law.
- *Concentration profile* is represented as a plot of concentration versus distance into the solid material.
- *Concentration gradient* is the slope of the concentration profile curve at some specific point.
- The *diffusion condition* for which the flux is independent of time is known as *steady state*.
- The driving force for *steady-state diffusion* is the concentration gradient (dC/dx).

Summary

- For **nonsteady-state diffusion**, there is a net accumulation or depletion of diffusing species, and the **flux is dependent on time**.
- The mathematics for **nonsteady state** in a single (x) direction (and when the diffusion coefficient is independent of concentration) may be described by *Fick's second law*.
- For a constant surface composition boundary condition, the solution to *Fick's second law* involves the **Gaussian error function**.
- The magnitude of the diffusion coefficient is indicative of the rate of atomic motion and depends on both host and diffusing species as well as on temperature.
- The **diffusion coefficient** is a function of temperature.
- The **two heat treatments** that are used to diffuse impurities into silicon during integrated circuit fabrication are **predeposition** and **drive-in diffusion**.

Summary

- **Diffusion faster** for; open crystal structures, materials with secondary bonding, smaller diffusing atoms and lower density materials.
- **Diffusion slower** for; close-packed structures, materials with covalent bond, large diffusing atoms and higher density materials.

<i>Equation</i>	<i>Solving For</i>	<i>Page Number</i>
$J = \frac{M}{At}$	Diffusion flux	143
$J = -D \frac{dC}{dx}$	Fick's first law	143
$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}$	Fick's second law	145
$\frac{C_x - C_0}{C_s - C_0} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$	Solution to Fick's second law—for constant surface composition	146
$D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$	Temperature dependence of diffusion coefficient	149

Thank you!